SIMATS ENGINEERING

ASSESSMENT TOOL 3

Comparative Analysis of Network Topologies

Annexure A

Course Name	Computer Networks
Course Code	CSA07
Course Outcome (CO4)	Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards (BL2, BL6)
Assessment Tool Weightage	30%
Total Marks	25
Number of Questions	5

Code of Conduct

I _Sk.SAMEER(192312356)(Name / Registration Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SK.SAMEER

Introduction

Network topology defines the logical and physical arrangement of nodes, links, and devices within a communication network, and directly influences cost, reliability, performance, scalability, and maintainability. This report presents a structured comparative analysis of four fundamental topologies — star, bus, mesh, and hybrid — against five carefully chosen criteria: cost, reliability and performance; fault tolerance; installation complexity; scalability; and maintenance. Each comparison is supported by a summary table, detailed technical reasoning, real-world implementation evidence, and a concluding insight, in line with the requirements of CO4 (design of network topologies using networking devices, transmission media, and cabling standards).

Question 1: Cost, Reliability, and Performance — Star vs. Bus Topology

Criteria Selected for Comparison

Cost (installation and cabling), reliability (impact of a single point of failure), and performance (behaviour under increasing network load) were selected because they represent the three factors most decisive to a network designer choosing between a centralised and a shared-medium topology.

Comparative Summary

Criterion	Star Topology	Bus Topology
Cost	Higher — requires a central switch/hub plus a dedicated cable run to every node	Lower — a single shared backbone cable serves all nodes
Cabling requirement	Point-to-point cable per node (more copper/fibre, more labour)	One continuous backbone with drop cables (minimal cabling)
Reliability	High — a faulty node or cable is isolated; the rest of the network is unaffected	Low — a break or fault in the backbone disables the entire segment
Performance under load	Stable — the central device manages traffic and limits collisions	Degrades — shared medium causes more collisions as devices are added

Detailed Analysis

Star topology's dependence on a central switch increases the bill of materials (active networking hardware) and cabling cost, since every device needs its own dedicated run back to that switch. Bus topology avoids this by tapping all devices onto one backbone, which is why it was historically favoured for small, budget-constrained LANs.

The cost advantage of bus topology, however, comes at the expense of reliability and performance. Because all nodes share one physical medium, a single cable break, a loose terminator, or a faulty tap can bring down communication for every connected device, and the shared-medium access method (CSMA/CD) causes throughput to fall sharply as more stations contend for the same channel. Star topology's central switch, in contrast, gives every node its own collision domain, so performance remains predictable even as the network grows, and a fault on one link never propagates to the rest of the network.

Real-World Evidence

Legacy 10BASE2/10BASE5 Ethernet installations used coaxial bus cabling for exactly this cost reason, but the industry moved almost entirely to switched star topology (10/100/1000BASE-T with structured cabling) once switch hardware became affordable, precisely because organisations needed the reliability and scalability that bus topology could not provide.

Conclusion & Insight

> *For small, temporary, or extremely cost-sensitive deployments, bus topology's low material cost can be attractive; but for any network where uptime and predictable performance matter — which includes almost all modern enterprise and campus LANs — star topology's higher upfront cost is justified by its superior fault isolation and consistent throughput.*

Question 2: Fault Tolerance — Mesh vs. Star Topology

Criteria Selected for Comparison

Fault tolerance was selected as the comparison criterion because it is the single most important factor for networks that must remain operational even when individual links or devices fail — a requirement typical of mission-critical and backbone infrastructure.

Comparative Summary

Criterion	Mesh Topology	Star Topology
Path redundancy	Multiple physical paths exist between any two nodes	Only one path exists — through the central hub/switch
Effect of a single link failure	Traffic automatically reroutes through an alternate path	Only the node on that link is affected; other nodes still work
Effect of a central-device failure	No single central device exists, so there is no equivalent single point of failure	Catastrophic — the entire network goes down if the hub/switch fails
Overall fault-tolerance level	Very high	Moderate

Detailed Analysis

In a full or partial mesh, every node connects to multiple other nodes, so the network has inherent path diversity. If one link or intermediate node fails, routing protocols can redirect traffic over a surviving path, meaning end-to-end communication is preserved even during multiple simultaneous failures in a rich mesh.

Star topology handles individual node or cable failures gracefully because each node has its own dedicated link, but it has one structural weakness: every path passes through the central switch or hub. If that device fails, the entire network segment loses connectivity at once, regardless of how healthy the individual nodes are. Mesh topology has no equivalent single point of failure, which is why it is rated more fault-tolerant overall despite being far more expensive to build.

Real-World Evidence

The core of the Internet, and large ISP backbones, are built as partial mesh networks of routers with multiple redundant links precisely so that a single fibre cut or router outage does not isolate entire regions. Military and disaster-recovery communication networks also favour mesh design for the same reason. Enterprise LANs, by contrast, typically accept star topology's moderate risk and mitigate the central-device weakness with a redundant, high-availability switch rather than a full mesh, since a full mesh at the desktop level would be economically impractical.

Conclusion & Insight

> *Mesh topology is unquestionably more robust for critical environments where continuous availability outweighs cost, whereas star topology remains the practical choice for everyday networks, provided the central device's reliability is reinforced through redundancy or high-quality hardware.*

Question 3: Installation Complexity — Bus vs. Mesh Topology

Criteria Selected for Comparison

Installation complexity was chosen because it directly determines deployment time, labour cost, and how well a topology scales physically as the number of devices increases — an essential consideration when selecting cabling standards for a new network.

Comparative Summary

Criterion	Bus Topology	Mesh Topology
Cabling required	One backbone cable with drop connections	$n(n-1)/2$ links for a full mesh of n nodes
Number of ports per device	One connection point (tap) per node	One port per connection — a node in a full mesh needs $(n-1)$ ports

Criterion	Bus Topology	Mesh Topology
Installation effort	Low — simple, linear layout	Very high — every node must be individually wired to every other node
Growth impact	Minor — new nodes tap onto the existing backbone	Severe — cabling and ports required grow quadratically with each new node

Detailed Analysis

Bus topology requires only a single backbone run, so installation is largely a matter of laying one cable and attaching drop connectors for each device — a process that scales linearly and remains manageable even for technicians with limited networking expertise.

Mesh topology, by contrast, requires a dedicated physical link between every pair of nodes. For a network of n devices, a full mesh needs $n(n - 1)/2$ individual cables, so the cabling and port count grow quadratically rather than linearly. Adding just a few extra nodes to a mesh can multiply the required wiring, making installation time-consuming, expensive, and difficult to manage neatly, particularly with copper or fibre cabling standards that impose distance and bend-radius constraints.

Real-World Evidence

This is why full mesh wiring is rarely used at the LAN edge and is instead reserved for a small number of high-value core devices — for example, a handful of core routers in a data centre interconnected in a mesh — where the number of nodes is kept deliberately small so the cabling stays manageable, while bus-style or star-style structured cabling is used everywhere else.

Conclusion & Insight

> *Bus topology is clearly easier and faster to install for small or simple networks, while mesh topology's installation burden makes it suitable only where the small number of highly critical nodes justifies the quadratic growth in cabling and cost.*

Question 4: Scalability — Hybrid vs. Star Topology

Criteria Selected for Comparison

Scalability was selected because it reflects how easily a network can grow to meet increasing organisational demand — a key factor when designing topologies intended to serve an expanding enterprise or campus.

Comparative Summary

Criterion	Hybrid Topology	Star Topology
Basis of design	Combines two or more topologies (e.g., star clusters linked by a mesh or bus backbone)	Single central device connects all nodes
Expansion method	New segments/topologies added without disturbing the existing structure	New nodes added only until the central device's ports are exhausted
Limiting factor	Practically none — architecture is designed to expand	Capacity of the central switch/hub
Suitability for growth	Excellent — ideal for large, evolving networks	Good, but limited beyond the central device's capacity

Detailed Analysis

Hybrid topology is built specifically to combine the strengths of multiple topologies — for instance, several star-wired departmental LANs connected through a mesh or bus backbone. Because each segment is semi-independent, new departments, buildings, or device clusters can be added as additional star segments without redesigning or disrupting the network that already exists.

Star topology can also scale, but only within the physical and logical limits of its central device: once every port on the switch or hub is occupied, further growth requires purchasing and integrating additional hardware, and cascading multiple switches introduces its own complexity and potential bottlenecks. Hybrid topology avoids this ceiling because growth is handled by adding entire new segments rather than by exhausting a single device's capacity.

Real-World Evidence

Large university campuses and enterprises commonly deploy exactly this hybrid pattern: individual buildings use star-wired Ethernet to connect end devices to a local switch, while a mesh or high-speed bus backbone interconnects the building switches, allowing the institution to add new buildings or departments over time without re-engineering the entire network.

Conclusion & Insight

> *Hybrid topology offers substantially greater long-term scalability than star topology because it grows by adding structured segments rather than by pushing a single central device to its limit, making it the preferred choice for large or continually expanding organisations.*

Question 5: Maintenance Complexity — Across Star, Mesh, Bus, and Hybrid Topologies

Criteria Selected for Comparison

Maintenance complexity was chosen as the final criterion because it determines the long-term operational cost of a network — how quickly faults can be identified, isolated, and repaired after installation is complete.

Comparative Summary

Topology	Fault Isolation	Overall Maintenance Difficulty
Star	Easy — problems are traced directly to a specific node or cable run	Low
Bus	Difficult — a backbone fault can affect the whole segment and is hard to pinpoint	High
Mesh	Complex — the large number of interconnections makes tracing faults time-consuming	High
Hybrid	Variable — depends on which combined topology is affected	Moderate

Detailed Analysis

Star topology is the easiest to maintain because every node has its own dedicated cable back to the central device; a fault therefore points immediately to one specific link or node, letting a technician isolate and resolve it without disturbing the rest of the network.

Bus topology is harder to maintain than star topology because the backbone cable is shared: a break or a bad terminator anywhere along its length can disrupt the whole segment, and locating the exact fault point along a long, continuous cable run is often slow and labour-intensive. Mesh topology, despite being the most fault-tolerant in operation, is also difficult to maintain — its large number of interconnecting links makes documentation, cable tracing, and fault diagnosis considerably more complex as the network grows. Hybrid topology sits in between: routine maintenance within a single star segment is straightforward, but administrators must also understand and manage the backbone technology connecting the segments, which requires broader skill and careful monitoring, giving it a moderate overall maintenance burden.

Real-World Evidence

Network operations teams typically favour structured, star-wired access layers for exactly this reason — a technician can identify a faulty port or cable within minutes — while backbone links (bus or mesh) are given extra redundancy and closer monitoring precisely because faults there are harder to isolate and more disruptive when they occur.

Conclusion & Insight

> *Maintenance effort ranks from easiest to hardest as star, then hybrid, then bus and mesh, meaning organisations should weigh the operational cost of troubleshooting — not just the initial fault tolerance or installation cost — when selecting a topology for long-term deployment.*

Overall Conclusion

No single topology is universally superior; each represents a different trade-off among cost, reliability, performance, fault tolerance, installation effort, scalability, and maintainability. Bus topology minimises cost and installation effort but sacrifices reliability and is hard to maintain. Star topology offers the best balance of manageability, fault isolation, and moderate scalability, making it the default choice for most modern LANs. Mesh topology delivers the highest possible fault tolerance and is indispensable for critical backbone and core infrastructure, but its installation and maintenance costs grow quadratically with network size. Hybrid topology, by combining these designs, achieves the best overall scalability and is the architecture of choice for large, evolving enterprise and campus networks. Selecting an appropriate topology therefore requires weighing organisational priorities — budget, criticality of uptime, expected growth, and available technical support — against the strengths and weaknesses identified in this comparative analysis.